

How Concentrated is the SP500, really?

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Abstract

U.S. equity indices have become increasingly dominated by a small cluster of mega-cap firms, raising a practical question for risk management: when does index dominance translate into fragility, particularly in tails and drawdowns? To answer this we propose a *return-space dominance* index: the fraction of index variance explained by a small, economically interpretable driver set.

We formalize this return-space concentration index C_t as the rolling R^2 from regressing index returns on an equal-weight mega-cap basket. Within a factor-model framework, we show how rising name concentration can be variance-neutral, even as return-space dominance increases through time-varying exposures and correlated residuals. Empirically, using daily U.S. equity data since the early-2000s, we document historically elevated return-space dominance in the S&P 500 proxy (SPY). We also show that high- C_t regimes are associated with materially worse left-tail outcomes and deeper drawdowns with evidence indicating that the channel operates primarily through loss severity on down days rather than the frequency of down days.

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1 Introduction

Over the last decade, U.S. equity benchmarks have become increasingly dominated by a small cluster of mega-cap technology and technology-adjacent firms. These firms now account for a historically large share of index capitalization and, more importantly, for a disproportionate share of realized index returns. This raises a natural risk question: when does index dominance translate into fragility, particularly in left tails and drawdowns?

A central theme of this paper is that “concentration” is not a simple scalar object. Weight-based measures such as the Herfindahl–Hirschman Index (HHI) summarize capital allocation, but portfolio risk is governed by factor exposures, correlation structure, and state dependence. As a result, changes in name concentration need not map cleanly into changes in volatility or tail risk.

We therefore focus on *return-space concentration*: the degree to which a small driver set explains realized index variance. We construct a return-space concentration index C_t as the rolling R^2 from regressing SPY returns on an equal-weight mega-cap basket. This object directly measures dominance in realized returns rather than dominance in weights.

Core claim. Concentration risk is fundamentally a property of return-space dominance and conditional tail behavior, not of weights alone. Weight concentration can be variance-neutral, while return-space concentration can rise sharply and alter tail risk through correlated exposures and residual structure.

Contributions. This paper makes three contributions. First, we formalize the distinction between name concentration and return-space concentration and explain why weight-based measures are an imperfect proxy for risk. Second, within a factor-model framework, we derive conditions under which rising name concentration is variance-neutral and show that residual risk depends on implementation even when factor exposures are fixed. Third, using U.S. equity data, we document historically high return-space dominance in the S&P 500 and show that elevated dominance is associated with materially worse tail outcomes.

2 Related literature

2.1 Concentration, diversification, and portfolio structure

A long literature studies the relationship between portfolio concentration and risk. Early work emphasizes that diversification benefits depend on correlation structure rather than the number of holdings alone [Roll, 1992, Campbell et al., 2001]. Empirical studies document substantial heterogeneity in diversification outcomes even for portfolios with similar effective numbers of assets [Goetzmann and Kumar, 2008]. From an optimization perspective, simple diversification rules can outperform mean–variance portfolios in the presence of estimation error [DeMiguel et al., 2009], highlighting that concentration metrics based on weights alone are insufficient statistics for risk.

More recent work documents that rising concentration in equity indices does not mechanically translate into higher volatility or worse average performance, underscoring the importance of factor exposures and residual risk structure [Kritzman and Turkington, 2025, Bessembinder, 2018].

2.2 Factor structure, common variation, and dominance

The idea that a small number of common components can explain a large fraction of asset return variation is central to modern asset pricing. Approximate factor models show that aggregate risk is driven by dominant eigenmodes of the covariance matrix rather than by individual assets [Chamberlain and Rothschild, 1983, Connor and Korajczyk, 1986]. Empirical evidence confirms that markets periodically enter regimes of elevated commonality, in which correlations rise and diversification benefits deteriorate [Bai and Ng, 2002, Pukthuanthong and Roll, 2009].

Random matrix theory further demonstrates that changes in the leading eigenvalues of the covariance matrix can materially affect portfolio risk even when marginal volatilities are stable [Laloux et al., 1999, Plerou et al., 1999, Ledoit and Wolf, 2004]. These insights motivate return-space diagnostics that measure dominance in realized variance rather than capital allocation.

2.3 Tails and draw downs

A separate literature emphasizes that financial risk is inherently non-Gaussian and state dependent. Asset returns exhibit heavy tails, volatility clustering, and draw downs that are not well captured by second moments alone [Mandelbrot, 1963, Embrechts et al., 1997, Cont, 2001]. Extreme events are often associated with amplification mechanisms tied to leverage, liquidity, and common exposures [Gabaix, 2011].

Valuation plays a central role in shaping tail outcomes. Periods of elevated prices relative to fundamentals are associated with increased crash risk and asymmetric downside behavior [Shiller, 2000, Barro, 2006, Daniel and Moskowitz, 2016]. Investor flows and sentiment further interact with market structure, producing feedback effects that can amplify drawdowns in dominant regimes [Greenwood and Shleifer, 2014, Gabaix and Koijen, 2021, Goldberg et al., 2023].

3 Name concentration vs return-space concentration

3.1 Name concentration: HHI and effective number

Let $w_i \geq 0$ denote the index weight of firm i with $\sum_{i=1}^N w_i = 1$. The Herfindahl–Hirschman Index (HHI) is

$$\text{HHI} = \sum_{i=1}^N w_i^2, \quad N_{\text{eff}} = \frac{1}{\text{HHI}}. \quad (1)$$

These are purely functions of weights; they do not incorporate volatilities, correlations, factor exposures, or flow impact. Two portfolios may have identical N_{eff} yet radically different risk profiles.

3.2 Three lenses: capital, activity, and risk concentration

We distinguish three mathematically distinct notions of concentration: (i) *capital concentration* (weights), (ii) *activity concentration* (dollar-volatility proxies), and (iii) *risk concentration* (variance contributions). These notions coincide only under restrictive conditions.

Capital concentration. This is the HHI/effective-number notion in (1).

Activity concentration (\$-vol). Let σ_i denote the (annualized) volatility of asset i and define activity weights

$$\tilde{w}_i := \frac{w_i \sigma_i}{\sum_{j=1}^N w_j \sigma_j}. \quad (2)$$

The associated activity effective number is

$$N_{\text{eff}}^{\$} := \frac{1}{\sum_{i=1}^N \tilde{w}_i^2}. \quad (3)$$

Risk concentration (variance-share). Let Σ denote the return covariance. Define ex ante risk shares

$$p_i := \frac{w_i(\Sigma w)_i}{w^\top \Sigma w}, \quad \sum_{i=1}^N p_i = 1, \quad (4)$$

and the risk effective number

$$N_{\text{eff}}^{\text{risk}} := \frac{1}{\sum_{i=1}^N p_i^2}. \quad (5)$$

A market can look “diversified” under one N_{eff} while being concentrated under another or vice versa. This paper’s $\mathcal{C}_t$ complements these ex ante objects by measuring *realized* dominance in return space.

3.3 Return-space concentration

Let $r_m(t)$ be index returns and let $r_d(t)$ be a return series capturing a driver set (e.g. an equal-weight mega-cap basket). Over a window W_t ending at t , define the rolling regression

$$r_m(u) = \alpha_t + \beta_t r_d(u) + \eta_t(u), \quad u \in W_t. \quad (6)$$

Definition 1 (Return-space concentration index). The return-space concentration index is

$$\mathcal{C}_t := R_t^2(r_m \sim r_d), \quad (7)$$

the coefficient of determination from (6) computed on W_t .

$\mathcal{C}_t$ measures the fraction of index variance explained by the driver set within the rolling window, i.e. concentration in *realized return variation*.

3.4 Why $\mathcal{C}_t$ is closer to risk than HHI

HHI is a statement about *weights*. Risk is a statement about the *distribution* of returns and the system’s response to shocks. Return-space dominance is a mechanism by which a small cluster can drive index outcomes even if weights-based measures are ambiguous (e.g., large firms are diversified yet share common growth/AI duration exposure).

In addition, a direct group-level measure like $\mathcal{C}_t$ comes without “attribution” intuition traps. For example, Euler contributions like the ones in the activity concentration (dollar vol) metric are additive, but when constituents are highly correlated, individual attributions can appear counterintuitive and are sensitive to covariance shifts. The regression R^2

captures the *net* explanatory power of the group as a whole, aligning with the question “how much of the index is essentially this cohort?”

Furthermore, risk-contribution measures like the variance-share risk contribution (that we would classify as a far second after C_t) rely on Σw and therefore inherit sensitivity to covariance estimation choices (window length, shrinkage, missing data, microstructure effects); thus, they can be more useful when they are ex ante calculated within the context of a certain risk model and forecasted covariance. Whereas, by definition, C_t is an ex post (backward-looking) metric which is an easily communicated *observed dominance* diagnostic.

Lastly, risk contributions require consistent covariance estimation and universe definitions. A single-factor regression R^2 is straightforward to compute across different indices and time periods, facilitating comparisons (e.g., across benchmarks, sectors, or regions) without extensive covariance-model plumbing.

4 A stylized theoretical framework

4.1 Factor model and neutrality conditions

Consider a linear factor model for N assets:

$$r(t) = Bf(t) + \varepsilon(t), \quad (8)$$

where $r(t) \in \mathbb{R}^N$ are excess returns, $f(t) \in \mathbb{R}^K$ are factors with covariance Σ_f , $B \in \mathbb{R}^{N \times K}$ is the loading matrix, and $\varepsilon(t)$ has covariance Σ_ε . For a portfolio $w \in \mathbb{R}^N$ with $w^\top \mathbf{1} = 1$,

$$\text{Var}(w^\top r) = w^\top B \Sigma_f B^\top w + w^\top \Sigma_\varepsilon w. \quad (9)$$

Thus, name concentration can be variance neutral if it leaves both factor exposure $B^\top w$ and residual aggregation $w^\top \Sigma_\varepsilon w$ roughly unchanged.

Same factor exposure implies that systematic (factor-driven) risk changes little. In particular, if we “shift toward” a small set of large firms whose factor loadings are similar to those already represented in the portfolio, then the portfolio’s aggregate factor exposure

$$\beta_p = B^\top w \quad (10)$$

remains approximately unchanged (i.e., $\beta_p \approx \beta_{p,0}$ relative to a baseline portfolio with more diffuse weights w_0). Put differently, the portfolio’s economic exposures are essentially stable, so the factor-driven variance $\beta_p^\top \Sigma_f \beta_p$ is largely unaffected.

Moreover, if the reweighting concentrates on firms with lower idiosyncratic variance, the residual component $w^\top \Sigma_\varepsilon w$ can decline enough to offset any increase in concentration. In that case, weight-based concentration may rise even as total portfolio variance remains approximately unchanged (“variance-neutral”).

4.2 Implementation risk: fixed exposure does not fix residual risk

Even holding factor exposures fixed, the residual term in (9) can change across implementations. This result formalizes why “equal factor exposure” does not imply equal risk: residual covariance and implementation choices can dominate, which is precisely why HHI-based tests may lack power.

Theorem 1 (Unique minimum idiosyncratic-risk implementation). *Assume $\Sigma_\varepsilon \succ 0$. Fix a target factor exposure $\beta_\star \in \mathbb{R}^K$ and consider the feasible set*

$$\mathcal{W} = \{w \in \mathbb{R}^N : w^\top \mathbf{1} = 1, B^\top w = \beta_\star\}. \quad (11)$$

Then there exists a unique $w^\dagger \in \mathcal{W}$ minimizing residual variance $w^\top \Sigma_\varepsilon w$ over $\mathcal{W}$. Moreover, for any $w \in \mathcal{W}$, $w \neq w^\dagger$, we have $w^\top \Sigma_\varepsilon w > (w^\dagger)^\top \Sigma_\varepsilon w^\dagger$.

Remark 1. Theorem 1 clarifies why HHI-based tests can be underpowered: a portfolio can have unchanged HHI yet higher residual risk due to correlated residuals or a worse implementation; conversely, HHI can rise while residual risk falls if idiosyncratic variances drop sufficiently. Return-space dominance (high $\mathcal{C}_t$) captures a different phenomenon: a larger fraction of realized variance is explained by the driver set.

4.3 Link to tails and fragility

Variance is not the whole story. If the driver set has non-linear exposures (e.g. long-duration growth, option-induced convexity) or if the market is inelastic, then return-space dominance can amplify left tails. A reduced-form implication is that the conditional distribution of r_m differs across regimes of $\mathcal{C}_t$, even if unconditional volatility is similar. Such mechanisms are consistent with theories of endogenous risk amplification and granular shocks in dominant firms [Gabaix, 2011].

5 Empirical design

5.1 Data & Portfolios

We use daily adjusted prices and analyze (i) index proxies SPY (S&P 500), RSP (S&P 500 equal-weight), QQQ (Nasdaq 100), IWM (Russell 2000), and (ii) two equal-weight baskets: *M7_equal* (AAPL, MSFT, AMZN, GOOG, META, NVDA, TSLA) and *Tech2000_equal* (MSFT, CSCO, INTC, ORCL, IBM, QCOM).

In what follows, stocks make it to our 2 baskets as soon as they go public. That is true for *M7_equal* as well: 5 out of the 7 stocks have been public effectively since our starting date, and all have been trading publicly for at least the last 12 years or so. Similarly, most have had at least 12 years of SPY membership, with the exception of TSLA that was included in SPY end of 2020.

For this portfolio, we have also checked how things would look like if instead we used today’s relative weights, rather than equal weights. For all results we will be sharing, we did not derive any different conclusion. In addition, we see various practical reasons for using an equal weight basket:

1. Robustness to mechanical cap-weight drift: Cap weights drift continuously with prices, corporate actions, float adjustments, and reconstitutions. An equal-weight top- N basket is often more stable and less dominated by the top one or two names, making R^2 a cleaner time-series monitor of whether *the index co-moves with the dominant cohort* rather than merely reflecting the current weight distribution.
2. Equal-weighting tends to emphasize the *collective* movement of the dominant cohort (a “theme” or factor-like component) rather than the idiosyncratic influence of the very largest names. When stakeholders mean “concentration” as *one narrative driving returns*, a high R^2 is an interpretable indicator of one-factor dominance.

3. It offers an explainability KPI that is easy to communicate. Statements like “*the equal-weight top 10 explains 85% of index variance*” are easily understood by non-technical audiences. As already discussed, risk contributions are economically meaningful, but can be harder to explain succinctly because they involve marginal contributions and covariance structure.

In summary, we are not looking to do an economic variance attribution. Instead, we focus on producing a *practical, stable, and comparable* diagnostic of *common-factor dominance* of the dominant cohort.

5.2 Key Statistics, Calculations and Experiments

We compute:

- Long-run performance and risk metrics (annualized return/volatility, Sharpe ratio, max draw down (*MDD*), historical VaR/CVaR);
- Rolling 1-year volatility for all series;
- Rolling regression $SPY \sim M7_equal \beta_t$ and $C_t = R_t^2$;
- Volatility decomposition of SPY into ”factor-driven” (*M7_equal*) and residual components;
- Regime comparisons of tails for SPY conditional on C_t (low/medium/high by terciles);
- Present and historical statistics on C_t
- A diagnostic dynamic allocation rule mapping C_t to an equity weight;
- Downside regression of SPY returns on C_t

6 Results

6.1 Long-run performance

We show the long term statistics for the various indices and our 2 baskets in Table 1. In addition to return and volatility, we provide tail risk metrics such as MDD, Value at Risk (VaR) and Conditional Value at Risk (CVaR). We compare the first 2 moments of returns for all portfolios in Figure 1 where we show the growth of 1USD and the one-year rolling annualized volatility, in panels (a) and (b), respectively.

Table 1: Long-run performance and risk metrics (in %) of the indices and baskets considered in the period 05/01/2003-12/31/2025. Results are obtained using daily returns and are annualized (252 trading days/year).

	Return	Vol	Sharpe	MDD	VaR _{5%}	CVaR _{5%}	VaR _{1%}	CVaR _{1%}
SPY	11.3	18.6	0.61	-55.2	-1.7	-2.8	-3.4	-5.0
RSP	11.1	19.9	0.56	-59.9	-1.8	-3.0	-3.6	-5.3
QQQ	15.6	21.5	0.72	-53.4	-2.2	-3.2	-3.8	-5.0
IWM	10.0	23.8	0.42	-58.6	-2.3	-3.5	-3.9	-5.8
M7_equal	34.1	28.3	1.21	-63.9	-2.8	-4.1	-4.8	-6.2
Tech2000_equal	13.6	22.3	0.61	-42.8	-2.1	-3.2	-3.7	-5.1

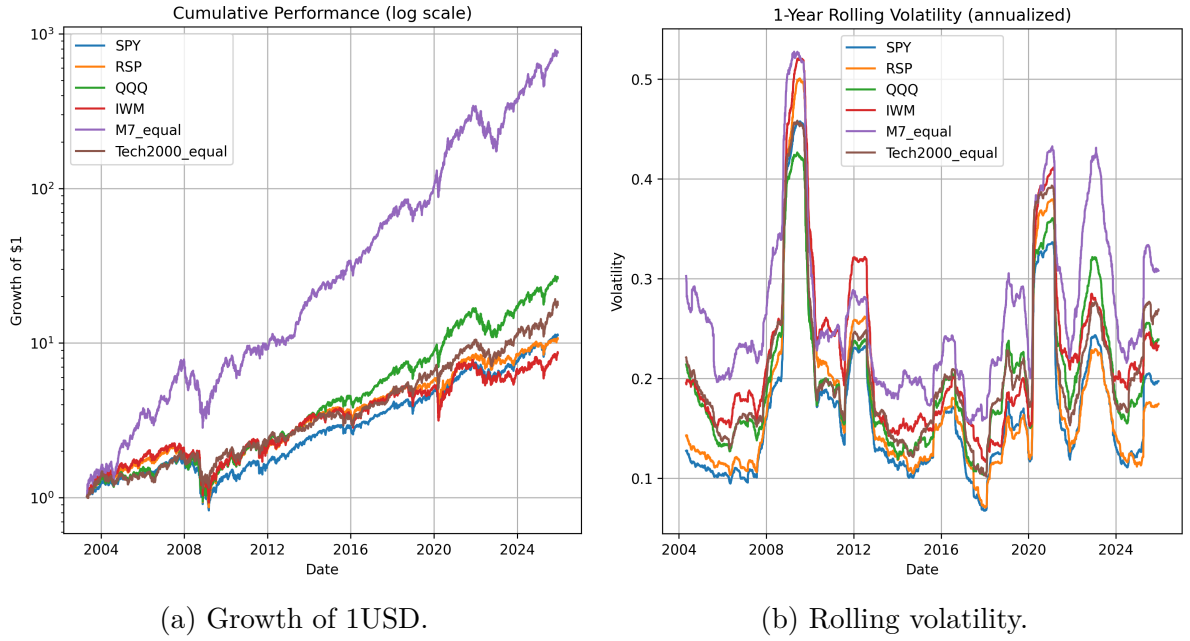
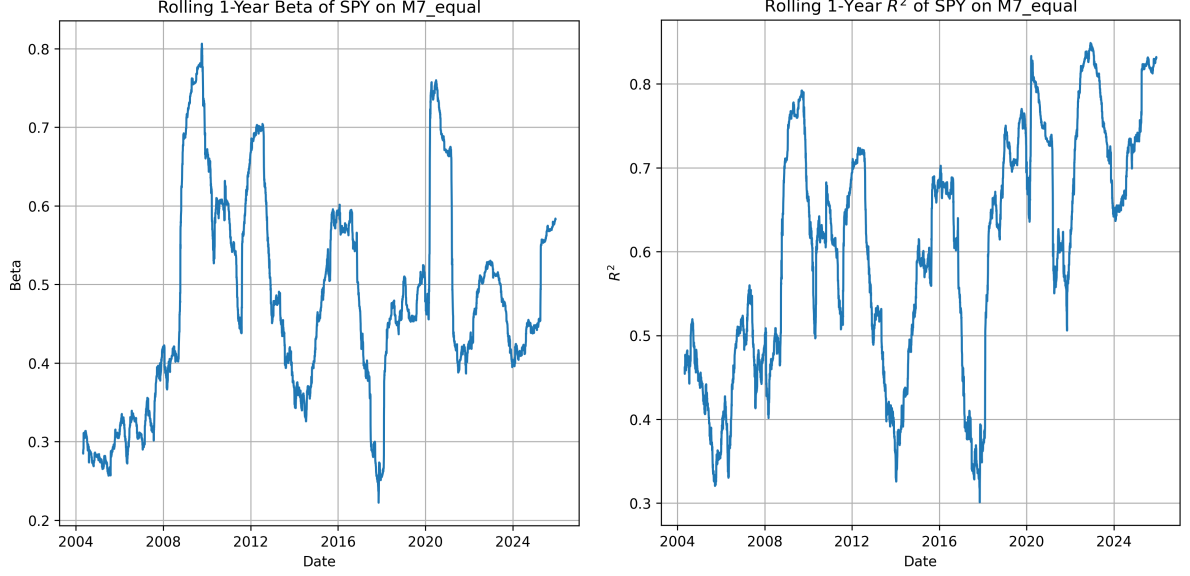


Figure 1: Cumulative performance (log scale) and 1-year rolling annualized volatility for SPY, RSP, QQQ, IWM, *M7_equal*, and Tech2000.

6.2 Return-space concentration for SPY

6.2.1 Variance decomposition



(a) Rolling beta.

(b) Rolling R^2 (return-space concentration).

Figure 2: Rolling 1-year regression of SPY on $M7_equal$.

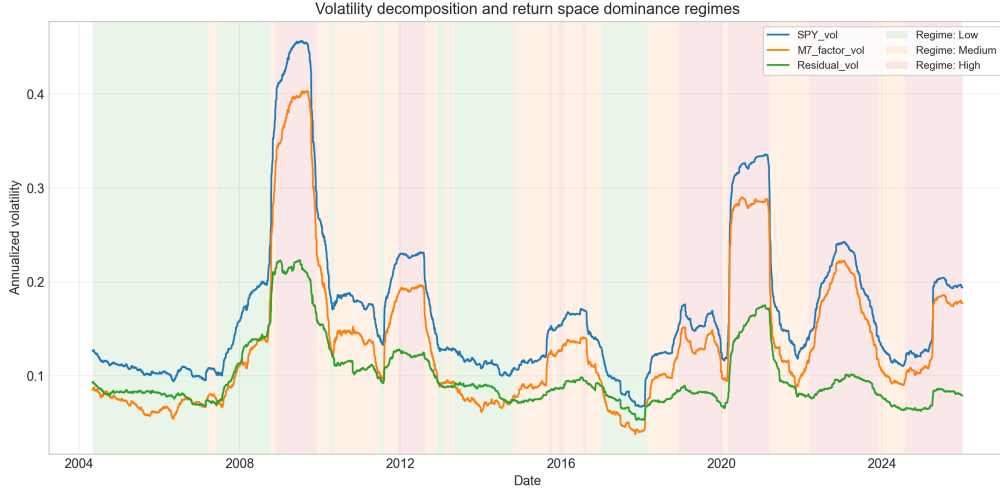


Figure 3: Rolling 1-year volatility decomposition of SPY into $M7_equal$ -driven factor volatility and residual volatility.

We show the β and C_t from rolling regressions in Figure 2 - left and right hand panel, respectively. Of course, the two metrics relate - they differ in that β is a rescaled $\sqrt{C_t}$, with the rescaling being how much more volatile SPY is from . Thus, even though C_t is unitless (/normalized) and gives the % variance in the SPY, explained by $M7_equal$, β is connecting the change in SPY for a unit change in $M7_equal$. We show the rolling 1-year volatility decomposition of the SPY into -driven factor volatility and residual volatility in Figure 3.

We interpret C_t as the fraction of SPY variance explained by the mega-cap driver set. High C_t indicates that index variance is increasingly driven by the mega-cap basket. We use the equal-weight basket as a transparent, investable proxy for the dominant leader cohort; however, qualitatively similar dominance dynamics are obtained with alternative

top- k mega-cap driver sets, though magnitudes vary with the definition of the driver universe.

6.2.2 Regimes and Tails

We use C_t to identify different regimes, based on the return space dominance. Specifically, we look into three regimes, aligned with C_t terciles - top, bottom and middle 1/3 of return space concentration. The regimes are shown in Figure 3 with different shade colors. With the exception of a few peaks around 2016, our concentration metric is stable, characterizing indeed regimes, i.e., contiguous chunks of time that also belong together from the dynamics point of view.

This volatility and regime decomposition clarifies why HHI regressions can miss economically meaningful concentration: concentration in *return-space* can rise even when total volatility is stable if residual risk falls or if exposures rotate.

For each regime, we provide tail risk metrics in Table 2. A key implication is that tail diagnostics can change substantially across regimes even when unconditional volatility does not, consistent with a fragility channel.¹

Table 2: SPY tail risk (in %) by return-space concentration regime.

	VaR _{5%}	CVaR _{5%}	VaR _{1%}	CVaR _{1%}
Low	-1.4	-2.2	-2.5	-3.5
Medium	-1.8	-3.0	-3.6	-5.4
High	-2.1	-3.3	-4.2	-5.6

6.2.3 How concentrated is SP500, really?

We explained the rationale behind our choice of the $M7_equal$, as part of a broader effort to assess whether the current period is exceptional in terms of return-space concentration. However, before using C_t to answer this question, we have to make sure that C_t via $M7_equal$ can indeed answer it. That is why we looked at how different things would look like for SPY C_t -wise, if instead we regressed using equal weight portfolios of 7 at random chosen components of the SP500 (the way it is today). We show the cross-sectional results for C_t in comparison to the C_t obtained by $M7_equal$ in Figure 4 and we show the present as well as the pooled distribution of C_t derived this way in the left and right hand panel of Figure 5, respectively.

¹This is an ex post, descriptive analysis. We examined several sensible implementation choices—e.g., defining C_t terciles using expanding rather than full-sample cutoffs, classifying regimes at time t using C_{t-1} versus C_t , and using block-bootstrap resampling to gauge sampling variability and the significance of cross-regime differences. That said, we make no forecasting claims. Ultimately, the statistics we report are realized, historical outcomes. In that spirit, a historical point estimate such as $CVaR_{1\%} = -3.5\%$ is economically—and practically—distinct from $CVaR_{1\%} = -5.6\%$, without the need for formal significance testing.

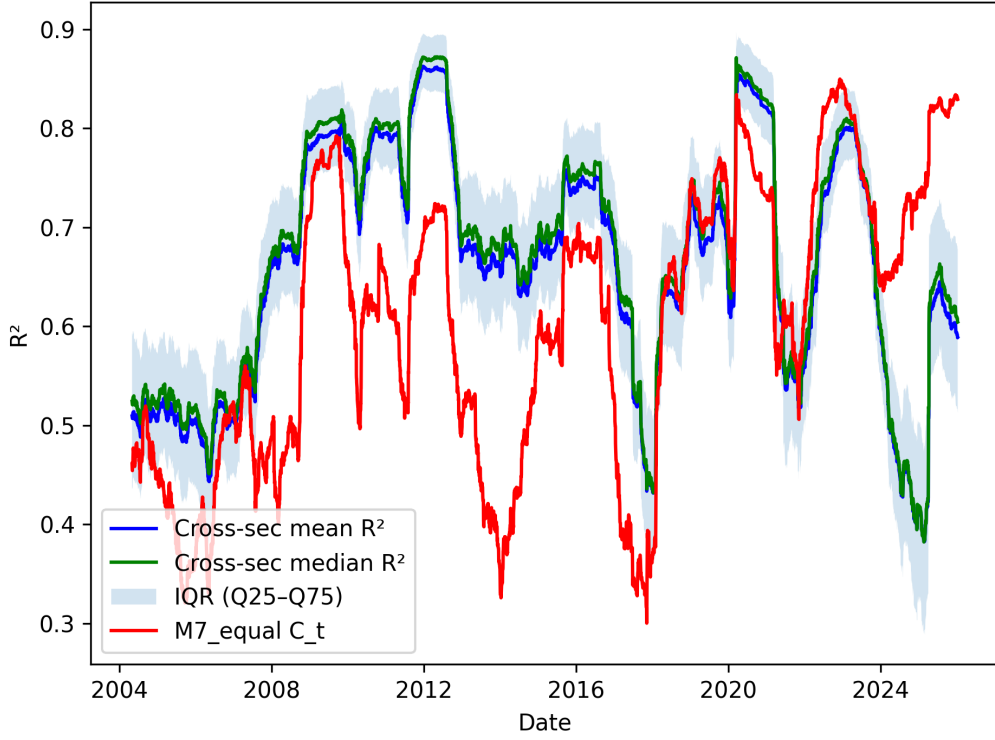


Figure 4: All results, except from the ones for $M7_equal$ are obtained by creating equal weight portfolios of 7 randomly chosen SP500 components (components as of today). The equally weighted basket of the Mag 7, that we use to define our SPY return space concentration, C_t , is shown in red.

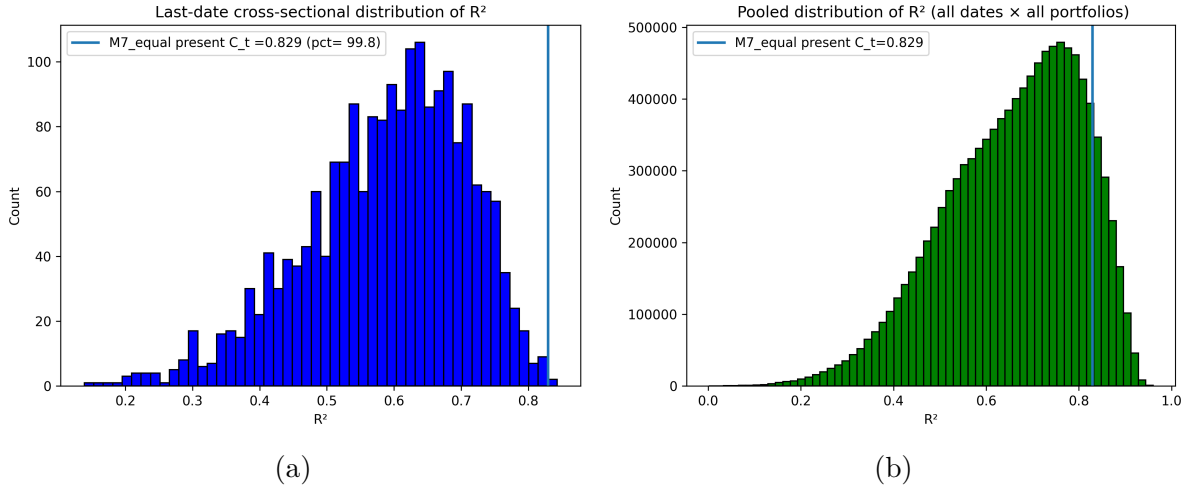


Figure 5: Rolling-window R^2 distributions from regressing SPY returns on equal-weight portfolios of 7 randomly selected SPY constituents. Using a 252-trading-day window, the left panel shows the cross-sectional distribution of R^2 across 2,000 random portfolios at the most recent date. The right panel shows the pooled distribution of R^2 values across all dates since mid-2003 (again from 2,000 portfolios). The vertical blue line marks the current R^2 of the equal-weight $M7_equal$ portfolio.

The present $M7_equal$ C_t is ~ 0.83 and that makes it higher than 99.8% of the C_t we get with any other randomly generated equal weight portfolios with 7 of the current SPY stocks and 88.8% of the C_t of such portfolios but when pooled across time (and since the start of our data). These statistics would indicate that the present is quite special in terms of the SPY return space concentration.

More results on the distributions of R^2 's obtained are shown in Table 3.

Table 3: Cross-sectional and pooled summaries of rolling R^2 (SPY on equal-weighted randomly selected 7-stock portfolios), with $M7_equal$ benchmark.

Statistic	Last-date	Pooled
Mean	0.59	0.66
Median	0.61	0.68
5th perc.	0.37	0.39
25th perc.	0.52	0.56
75th perc.	0.69	0.78
95th perc.	0.76	0.86
IQR	0.17	0.22
$M7_equal$ (present)	0.83	
$M7_equal$ perc. vs last-date dist.	99.75	
$M7_equal$ perc. vs pooled dist.	88.79	

6.3 Experiments

6.3.1 A diagnostic timing rule

We have seen that the present is pretty special in terms of C_t . Here we are trying to see whether it communicates something useful across time. We use the value of C_t to determine exposure on SPY. Specifically, after winsorizing C_t to ensure robustness against spikes and/or noise, and allowing for allocation weights from w_{min} to 1., we scale (linearly) the exposure, w , to SPY so as to reflect where C_t is in the C_{min}, C_{max} range, with lower w when C_t is higher.²

We show the allocation as well as the cumulative performance in Figure 6 and present core statistics for 3 allocation strategies in Table 4. Specifically, in addition to the SP500 and the dynamic strategy, we also show a static strategy that has a constant allocation equal to the average allocation of the dynamic strategy.

²We used $C_{min} = 0.2$, $C_{max} = 1$. and $w_{min} = 0.4$, $w_{max} = 1$.

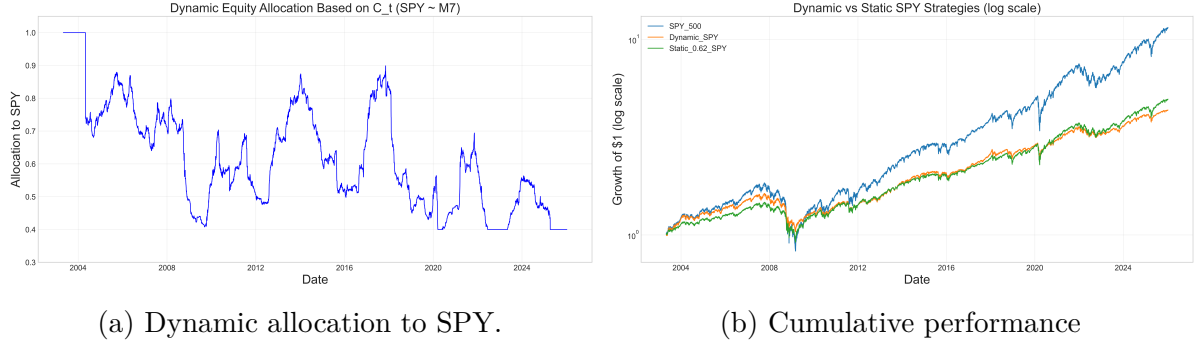


Figure 6: (a). Dynamic allocation to SPY as a function of rolling R^2 (return-space concentration). The more SPY variance is explained by , the more concentrated the index, and the lower the allocation to it. (b). Cumulative performance (log scale) of Dynamic_SPY, SPY_500, and Static_x_SPY (same average equity exposure as Dynamic_SPY).

Table 4: Performance of dynamic and static SPY allocation strategies. In the dynamic, the allocation scales inversely proportional to return space concentration. Even though all three strategies achieve on average the same return per unit risk (Sharpe), it is clear that tail risk is significantly avoided when we do not invest much on SPY in times of high return space concentration.

Portfolio	Return	Volatility	Sharpe	MDD
SPY_500	10.8	18.9	0.57	-55.2
Dynamic_SPY	6.6	11.9	0.56	-40.7
Static_0.68_SPY	7.7	12.9	0.60	-40.9

6.3.2 Downside Interaction and Channel Decomposition

We study how the concentration measure C_t relates to the downside behavior the SPY returns. Following a downside-focused specification, we define

$$y_t \equiv \min(r_t^{SPY}, 0), \quad (12)$$

so that $y_t = 0$ on up days and $y_t = r_t^{SPY} < 0$ on down days. We first estimate the baseline regression

$$y_t = \alpha + \beta C_t + \varepsilon_t, \quad (13)$$

by OLS. Since financial time series residuals commonly exhibit conditional heteroskedasticity and serial correlation, we report both conventional (homoskedastic) standard errors and heteroskedasticity-and-autocorrelation-consistent (HAC) Newey–West standard errors, using 5, 10 and 20 lags (i.e., allowing residual dependence up to that many trading days).³ We show the results of this downside regression in Table 5.

The estimated slope is negative and statistically significant under both inference schemes. Economically, a 0.10 increase in C_t corresponds to about -5 bps/day. The unconditional mean downside component is $\mathbb{E}[y_t]$ is about -36 bps/day, indicating that the effect size is meaningful relative to the typical downside contribution.

³HAC inference is particularly important here because y_t is truncated at zero and inherits time-varying volatility from r_t^{SPY} , both of which can induce heteroskedastic and autocorrelated regression errors.

Specification	$\hat{\beta}$	SE	t	p	CI Low	CI High
OLS (classic)	−0.0046	0.0007	−6.305	$< 10^{-6}$	−0.0060	−0.0032
OLS + HAC (NW,5)	−0.0046	0.0009	−5.376	$< 10^{-6}$	−0.0063	−0.0029
OLS + HAC (NW,10)	−0.0046	0.0010	−4.607	4.2×10^{-6}	−0.0065	−0.0026
OLS + HAC (NW,20)	−0.0046	0.0012	−3.908	9.4×10^{-5}	−0.0069	−0.0023
R^2	0.0072					

Table 5: Downside regression: $y_t = \min(r_t^{SPY}, 0)$ on concentration C_t

Channel	Spec.	$\hat{\beta}$	SE	t	p	CI Low	CI High
Frequency: $D_t = \mathbb{1}\{r_t < 0\}$	OLS	0.0176	0.0476	0.369	0.712	−0.0757	0.1109
	HAC (5)	0.0176	0.0456	0.385	0.700	−0.0719	0.1071
	HAC (10)	0.0176	0.0445	0.395	0.693	−0.0696	0.1048
	HAC (20)	0.0176	0.0452	0.389	0.697	−0.0711	0.1063
Severity: $r_t \mid (r_t < 0)$	OLS	−0.0100	0.0014	−7.213	$< 10^{-6}$	−0.0127	−0.0073
	HAC (5)	−0.0100	0.0020	−4.926	$< 10^{-6}$	−0.0139	−0.0060
	HAC (10)	−0.0100	0.0024	−4.132	3.7×10^{-5}	−0.0147	−0.0052
	HAC (20)	−0.0100	0.0028	−3.524	4.3×10^{-4}	−0.0155	−0.0044
R^2 (frequency)							0.0000
R^2 (severity)							0.0210

Table 6: Channel decomposition regressions: frequency vs. severity

Quantity	Value
$E[\min(r_t^{SPY}, 0)]$	−0.0036
$p = \Pr(r_t^{SPY} < 0)$	0.4445
$\mu_- = E[r_t^{SPY} \mid r_t^{SPY} < 0]$	−0.0080
β_f (frequency slope)	0.0176
β_s (severity slope)	−0.0100
$\mu_- \beta_f$ (frequency contrib.)	−0.0001
$p \beta_s$ (severity contrib.)	−0.0044
β_{downside} (decomp. sum)	−0.0045

Table 7: Moments and decomposition of the downside slope. Results are rounded to 4 decimal points (thus

To understand *which mechanism* drives the relationship in (13), we decompose the downside expectation into frequency and severity components. Using the identity

$$\mathbb{E}[\min(r_t, 0)] = \Pr(r_t < 0) \mathbb{E}[r_t \mid r_t < 0] \equiv p \mu_-, \quad (14)$$

a more negative downside component can arise from (i) a higher probability of down days (p increases) and/or (ii) larger losses conditional on down days (μ_- becomes more negative). We therefore estimate two auxiliary regressions:

Frequency channel. Define the down-day indicator $D_t \equiv \mathbb{1}\{r_t^{SPY} < 0\}$ and estimate

$$D_t = \alpha_f + \beta_f C_t + u_t. \quad (15)$$

In our sample, the frequency slope is small and statistically insignificant ($\hat{\beta}_f \approx 0.0176$, HAC $p \approx 0.7$), implying that changes in C_t do not meaningfully relate to the likelihood of a down day.

Severity channel. Conditioning on down days ($r_t^{SPY} < 0$), we estimate

$$r_t^{SPY} = \alpha_s + \beta_s C_t + v_t, \quad \text{for } r_t^{SPY} < 0, \quad (16)$$

which isolates the effect of C_t on the magnitude of losses. We find a negative and highly significant severity slope ($\hat{\beta}_s = -0.01$, HAC $p \lesssim 10^{-4}$), indicating that higher C_t is associated with materially worse returns *conditional* on being in a down market. For example, a 0.10 increase in C_t corresponds to roughly 10 bps lower average return on down days.⁴

Finally, combining (15)–(16) with (14) yields a first-order decomposition of the baseline slope:

$$\beta_{\text{downside}} \approx \mu_- \beta_f + p \beta_s. \quad (17)$$

Using the sample moments $p \approx 0.4445$ and $\mu_- \approx -0.0080$, the frequency contribution is $\mu_- \hat{\beta}_f \approx -0.00014$, whereas the severity contribution is $p \hat{\beta}_s \approx -0.00443$, summing to ≈ -0.0045 —very closely matching the baseline estimate $\hat{\beta}$. Hence, the observed relationship between concentration and the downside component of returns is driven almost entirely by the *severity* channel (about 97% of the total), rather than by changes in the frequency of down days. Importantly, this evidence pertains to downside risk and does not imply that unconditional mean returns decrease by the same amount; it reflects that losses are larger when they occur under high- C_t conditions.

7 Discussion and implications

This paper’s central message is that “concentration” is not adequately summarized by weights alone. In particular, the return-space concentration index C_t —implemented as the rolling R^2 from regressing SPY on an equal-weight mega-cap basket—directly measures how much of realized index variance is explained by a small, economically interpretable driver set. Empirically and conceptually, this object can move materially even when weight-based metrics are ambiguous.

7.1 Return-space dominance versus weight-based concentration

A key practical implication of the variance decomposition is that weight-based concentration (e.g., HHI) can miss economically meaningful changes in risk. The rolling volatility decomposition highlights that return-space concentration can rise even when total volatility is relatively stable—for example, when residual risk declines or when effective exposures

⁴Note that, for simplicity, we round results in this section to four decimal places. Even so, for our purposes—illustrating how strongly the severity of SPY down days depends on C_t —these figures should be interpreted as order-of-magnitude estimates rather than precise point values. Any apparent discrepancies are a consequence of rounding.

rotate. Put differently, a portfolio can become “more concentrated” in a capital-allocation sense without a commensurate increase in unconditional variance, yet still become more *fragile* if a larger fraction of the index’s realized variation is increasingly explained by a narrow driver cohort.

7.2 Regimes, tails, and the economics of fragility

Using C_t terciles to define dominance regimes yields a sharp empirical result: tail risk deteriorates monotonically with return-space dominance. This supports a “fragility” interpretation: when index outcomes are increasingly driven by a small cohort, adverse realizations for that cohort map more directly into index-level drawdowns.

An important nuance is that the downside relationship appears to operate primarily through the *severity* of down days rather than their *frequency*. The downside regression finds a statistically significant negative association between C_t and the downside component $y_t \equiv \min(r_t^{SPY}, 0)$, with economically meaningful effect sizes. The subsequent channel decomposition indicates that higher C_t does not meaningfully predict the likelihood of a down day, but it does predict larger conditional losses on down days (the severity channel). For risk management, this distinction matters: elevated dominance is less about “more frequent bad days” and more about “worse bad days when they arrive.”

7.3 Are we in an unusually concentrated market?

To benchmark the magnitude of current dominance, the paper compares the present C_t from the mega-cap basket to distributions obtained by regressing SPY on many random equal-weight 7-stock portfolios drawn from current SPY constituents. The present mega-cap C_t is approximately 0.83, placing it above 99.8% of the cross-sectional distribution at the last date and above 88.8% of the pooled distribution across time. These comparisons suggest that today’s return-space dominance is unusual in a historical sense, even relative to what one would obtain from arbitrary 7-stock cohorts.

7.4 A diagnostic allocation rule: what it shows (and what it does not)

The paper includes a simple diagnostic rule that scales SPY exposure inversely with C_t . The intent is not to claim a robust timing strategy, but to illustrate how return-space dominance could be mapped into an actionable “risk posture” knob. Consistent with the tail-risk evidence above, the diagnostic rule delivers materially smaller drawdowns than a static full-allocation benchmark in the historical sample, at the cost of lower average returns; yet, Sharpe ratios are similar. This reinforces two points: (i) concentration is a *risk characteristic* that can be managed, and (ii) reducing fragility is not “free alpha.”

7.5 Scope, robustness, and extensions

Several considerations shape how the results should be used in practice. First, the analysis is explicitly ex post and descriptive, and the paper does not make forecasting claims; multiple reasonable implementation checks are discussed, but the reported statistics remain realized historical outcomes. Second, while the equal-weight mega-cap basket is chosen for stability and interpretability, driver-set selection remains a modeling choice;

extensions could examine alternative top- k definitions, sector-neutral constructions, or factor-replicating portfolios. Third, the evidence points toward state-dependent channels (correlated exposures, residual structure) as mechanisms through which dominance alters tails rather than second moments alone; integrating explicit flow or liquidity measures would be a natural next step.

Overall, a practical implication is that concentration monitoring should be layered: weight-based metrics remain useful for describing capital allocation, but return-space dominance provides a complementary diagnostic that is closer to realized index behavior in stress-relevant states.

8 Conclusion

Weight concentration is an incomplete descriptor of index risk. The empirically relevant object for assessing fragility is return-space dominance: who explains index variance and how the conditional distribution of index returns changes when that dominance is elevated. Using a simple return-space concentration index based on rolling R^2 , we show that the S&P 500 exhibits historically high dominance by a small driver set (with current $C_t \approx 0.83$ ranking in the extreme right tail of comparable cohort-based benchmarks), and that high-dominance regimes are associated with materially worse left-tail outcomes. Moreover, the downside relationship operates mainly through loss severity rather than an increased frequency of down days, highlighting the importance of regime-aware tail diagnostics for risk management.

Taken together, these findings argue a comprehensive assessment of concentration risk requires diagnostics that live in return space and tails, complementing weight-based measures.

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